In Silico 4D-CT-Based Tumor Motion Modeling and Quantitative Evaluation of PTV Margins for Lung Radiotherapy Planning

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Abstract: Target motion during thoracic radiation therapy induces spatial misalignment between planned beam apertures and mobile tumor volumes. Current commercial treatment planning systems (TPS) utilize proprietary margin expansion modules that often treat intrafractional motion as an isotropic uncertainty. In this in silico study, we developed an open-source Python computational pipeline to track 3D tumor centroid trajectories across ten-phase 4D-CT volumes and quantify target coverage under uniform planning target volume (PTV) margin expansions using a synthetic digital thorax phantom. A calibrated spatiotemporal thorax tensor (T, Z, Y, X) with voxel spacing of 2.0 × 1.5 × 1.5 mm^3 was synthesized across respiratory phases T0 to T9. Solid tumor volumes were segmented within an anatomically isolated pulmonary region of interest via intensity thresholding (-300 to +150 HU) and tracked using spatial moments across regular, deep, and irregular breathing patterns. Empirical target coverage rates (CR) for margins of 2, 5, 8, 10, and 12 mm were benchmarked against the analytical van Herk formula ($2.5\Sigma + 0.7\sigma$) and ICRU 62/83 consensus criteria (CR ≥ 95%). Tumor excursion was dominated by craniocaudal displacement, with peak-to-peak amplitudes of 12.00 mm (regular), 22.00 mm (deep), and 14.58 mm (irregular), whereas lateral and anteroposterior excursions remained below 6.07 mm. Under regular respiration, a 12 mm margin achieved 98.4% minimum coverage. However, the standard 5 mm clinical margin exhibited severe geometric miss under deep (CR = 9.5%) and irregular breathing (CR = 52.6%). These quantitative findings confirm that uniform 3D margin expansions are suboptimal for tumors with craniocaudal excursions exceeding 10 mm, demonstrating the necessity of patient-specific anisotropic margins or active motion mitigation techniques consistent with AAPM Task Group 76 guidelines.

Keywords: 4D-CT, Lung Radiotherapy, Intrafraction Motion, In Silico Modeling, Centroid Tracking, PTV Margins, van Herk Recipe, AAPM TG-76.

1. Introduction

Geometric accuracy in thoracic external beam radiotherapy is fundamentally compromised by intrafractional respiratory motion. Contraction of the diaphragm and expansion of the thoracic cage induce substantial time-dependent displacements of lung parenchyma and embedded lesions. As detailed in the American Association of Physicists in Medicine (AAPM) Task Group 76 report [1], tumor motion exceeding 5 mm along any Cartesian axis demands explicit motion management to mitigate treatment-related toxicity and prevent tumor underdosing. In modern stereotactic body radiation therapy (SBRT), where fractional doses often exceed 10 Gy, unmanaged target excursion leads to steep dosimetric penalties at field boundaries [2, 3].

Conventional static three-dimensional computed tomography (3D-CT) acquisition captures arbitrary snapshots of breathing anatomy, resulting in motion-blurring artifacts, artificial volume elongation, and significant misestimation of gross tumor volume (GTV) boundaries [4, 5]. Four-dimensional computed tomography (4D-CT) resolves this temporal ambiguity by sorting projection data into discrete respiratory phase bins (typically 10 phases from peak-inhalation T0 to end-exhalation T5) correlated with an external respiratory surrogate [6, 7]. Advanced 4D-CT reconstruction methods, such as maximum intensity projection (MIP), allow the generation of an internal target volume (ITV) encompassing full respiratory excursion [8, 9].

Despite the clinical availability of 4D-CT, standardizing the planning target volume (PTV) margin from internal target boundaries remains challenging. According to ICRU Reports 62 and 83, the PTV must compensate for both internal

physiological variation and setup uncertainties to deliver at least 95% of the prescribed dose to the target [10, 11]. The analytical recipe developed by van Herk et al. ($M = 2.5\Sigma + 0.7\sigma$) provides a statistical population model ensuring 95% dose coverage in 90% of patients [12, 13]. Similarly, Stroom et al. proposed coverage probability formalisms for margin derivation [14]. However, commercial treatment planning systems (TPS) operate as closed, proprietary platforms, restricting clinicians from verifying how irregular breathing profiles, baseline shifts, and variable tidal volumes penalize empirical coverage across discrete phase bins [15, 16].

The primary objective of this in silico investigation is to construct a transparent, open-source computational pipeline in Python to: (1) synthesize 4D spatiotemporal anatomical volumes with calibrated attenuation coefficients; (2) extract 3D target centroids over ten breathing phases; (3) simulate kinematic responses to steady, deep, and irregular respiratory cycles; and (4) quantify geometric target coverage degradation across incremental margin expansions.

2. Materials and Methods

2.1. Spatiotemporal Phantom Formulation

An analytical 4D-CT thorax volume was formulated as a spatiotemporal discrete tensor $V(t, z, y, x) \in \blacksquare^{10 \times 60 \times 80 \times 80}$. Voxel dimensions were set to $\Delta z = 2.0$ mm, $\Delta y = 1.5$ mm, and $\Delta x = 1.5$ mm, representative of standard clinical thoracic helical CT protocols. Radiodensity was calibrated in standard Hounsfield Units (HU) using a linear attenuation mapping: $HU = I_{raw} \cdot s + b$, with slope $s = 1.0$ and intercept $b = -1024$. Anatomical compartments were assigned standard reference attenuation values: ambient surrounding air (-1000 HU), thoracic wall envelope (+30 HU), bilateral lung fields (-750 HU), and thoracic vertebral column (+650 HU). A spherical solid lesion of radius $R_0 = 12.0$ mm (+45 HU, baseline volume 7.24 cc) was positioned within the right lower lobe to reflect the anatomical region most vulnerable to diaphragmatic displacement.

2.2. Respiration Kinematics & Mathematical Parameterization

Tumor center displacement was modeled using the generalized periodic respiratory model described by Lujan et al. [17]:

$$\mathbf{r}_k(t) = r_{0,k} - A_k \cdot \cos^{2n}(\pi t/\tau - \phi_k) + \eta_k(t), \quad k \in \{X, Y, Z\},$$

where:

- $r_{0,k}$ represents the baseline Cartesian coordinate at end-exhalation (phase T5).
- $\mathbf{A} = (A_x, A_y, A_z)$ denotes the axis-specific peak-to-peak displacement amplitude vector. Because $\cos^{2n}(\cdot)$ ranges strictly between 0 and 1, the product $A_k \cdot \cos^{2n}$ traverses an excursion of exactly $A_k \cdot (1 - 0) = A_k$ across each breathing half-period. Thus, A_k mathematically specifies the full peak-to-peak displacement amplitude rather than the sinusoidal half-amplitude.
- $t \in \{0, 1, ..., 9\}$ denotes the discrete 4D-CT phase bin index representing phases T0 (peak inhalation) through T9 (late exhalation). The period is normalized to $\tau = 10$ phase bins, with phase $t = 10$ treated as cyclically equivalent to phase $t = 0$ (T0) to enforce periodic boundary closure across consecutive breathing cycles.
- $n = 1$ is the power coefficient reflecting the asymmetric physiological dwell ratio, modeling the extended resting phase during exhalation relative to inhalation (~60:40 dwell ratio).
- ϕ_k is the initial phase angle (set to 0 for synchronous displacement).
- $\eta_k(t)$ represents stochastic baseline drift and cycle-to-cycle amplitude variations.

Three distinct clinical breathing patterns were parameterized:

1. **Regular Respiration:** Synchronous harmonic breathing with nominal peak-to-peak amplitudes ($A_z = 12.0$ mm, $A_y = 3.5$ mm, $A_x = 1.2$ mm) and $\eta(t) = 0$.
2. **Deep Respiration:** Exaggerated tidal breathing modeling hyperventilation, with scaled peak-to-peak amplitudes ($A_z = 22.0$ mm, $A_y = 6.0$ mm, $A_x = 2.0$ mm).
3. **Irregular Respiration:** Tachypneic breathing with stochastic baseline fluctuations $\eta(t) \in [-1.8, +2.2]$ mm,

introducing cycle-to-cycle non-stationarity.

2.3. Spatial Moment Centroid Tracking & Pulmonary Region Isolation

Because chest wall musculature and mediastinal soft tissues exhibit radiodensities (+30 HU) that overlap with solid tumor attenuation (+45 HU), density thresholding [-300, +150 HU] was strictly applied within an anatomically isolated pulmonary region of interest (ROI) bounding the right lung field. Prior to tumor extraction, surrounding chest wall and bony structures were segmented out via automated lung parenchyma thresholding (-1000 to -500 HU), thereby preventing false-positive soft-tissue contamination. Within the pulmonary ROI, target voxels were extracted via density windowing [-300, +150 HU] followed by 3D binary morphological closing using a spherical structural element (kernel radius = 2 voxels).

The 3D spatial center-of-mass $C(t) = (Z_t, Y_t, X_t)$ at each phase t was computed from discrete spatial moments:

$$M_{pqr}(t) = \sum_z \sum_y \sum_x z^p \cdot y^q \cdot x^r \cdot M_t(z, y, x),$$

$$Z_t = \Delta z \cdot (M_{100} / M_{000}), \quad Y_t = \Delta y \cdot (M_{010} / M_{000}), \quad X_t = \Delta x \cdot (M_{001} / M_{000}),$$

where $M_t(z, y, x) \in \{0, 1\}$ is the binary tumor mask, and M_{000} represents total voxel count.

2.4. PTV Margin Construction & Coverage Metrics

The reference GTV was established at end-expiration (phase T5). Planning target volumes (PTV_m) for uniform margins $m \in \{2, 5, 8, 10, 12\}$ mm were constructed using exact 3D Euclidean Distance Transforms (EDT) via scipy.ndimage.distance_transform_edt. At each respiratory phase t, the target coverage rate CR(m, t) was quantified as:

$$CR(m, t) = (\sum_r [M_t(r) \wedge PTV_m(r)] / \sum_r M_t(r)) \times 100\%.$$

Compliance with ICRU Report 62/83 consensus requires that the minimum coverage across all ten phases satisfies:

$$CR_{min}(m) = \min_{t \in [0,9]} CR(m, t) \geq 95.0\%.$$

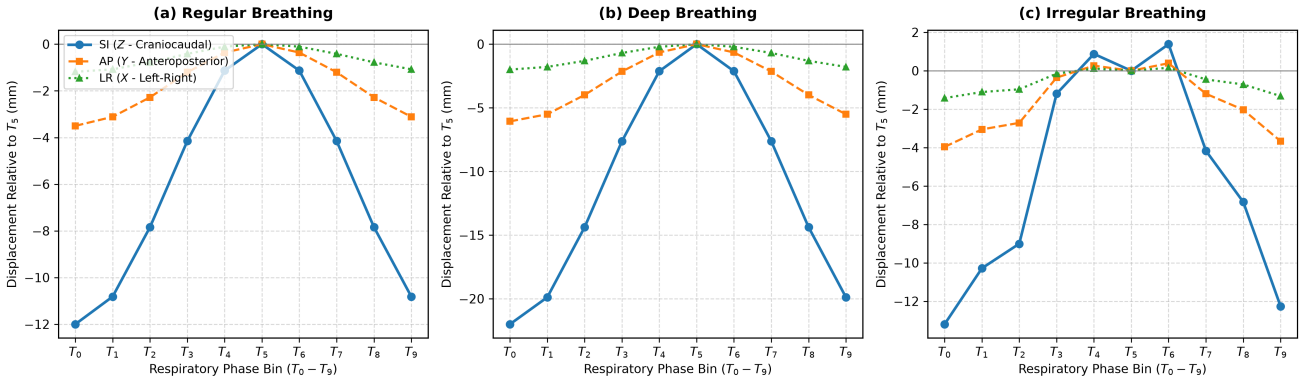


Figure 1. 3D Kinematic trajectory decomposition across 10 phases along SI, AP, and LR axes.

3. Results

Kinematic evaluation demonstrated that tumor excursion is overwhelmingly dominant along the craniocaudal (SI, Z) axis. Directional kinematic trajectory metrics are summarized in Table 1, and quantitative PTV coverage rates across margin expansions are detailed in Table 2.

Table 1. Directional kinematic displacement metrics across simulated breathing profiles.

Breathing Profile	LR (X) p2p	AP (Y) p2p	SI (Z) p2p	Max 3D Disp.
Regular Respiration	1.18 ± 0.42 mm	3.50 ± 1.22 mm	**12.00 ± 4.24 mm**	12.56 mm
Deep Respiration	2.00 ± 0.70 mm	6.07 ± 2.13 mm	**22.00 ± 7.77 mm**	22.91 mm
Irregular Respiration	1.58 ± 0.57 mm	4.36 ± 1.58 mm	**14.58 ± 5.30 mm**	13.85 mm

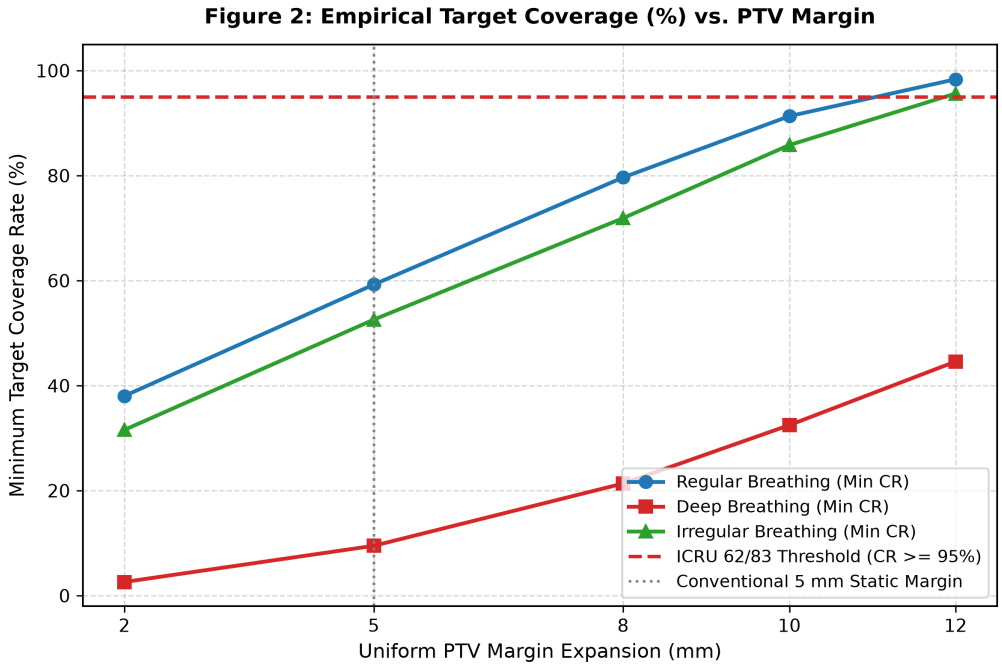


Figure 2. Empirical target coverage rate (%) as a function of PTV margin expansion.

Table 2. Empirical PTV target coverage rate (CR%) as a function of margin expansion.

** Note: Clinical acceptance is defined as $\min_t CR(m, t) \geq 95.0\%$ in accordance with ICRU Report 62/83 guidelines. Values in parentheses represent continuous analytical sphere volumes ($V = 4/3 \cdot \pi \cdot (R_0+m)^3$). Empirical PTV volumes are computed directly from voxel counts on the discrete 2.0 × 1.5 × 1.5 mm^3 anisotropic lattice (voxel volume = 4.50 mm^3) via 3D Euclidean distance transforms (EDT); slight differences from continuous volumes arise due to anisotropic spatial discretization and partial-volume boundary truncation.*

Margin	Volume (cc)*	Regular (Min CR)	Regular (Mean CR)	Deep (Min CR)	Irregular (Min CR)
2 mm	10.0 (11.5)	38.0%	72.7%	2.6%	31.6%
5 mm	19.2 (20.6)	59.3%	86.4%	9.5%	52.6%
8 mm	31.4 (33.5)	79.7%	95.1%	21.4%	71.9%
10 mm	42.2 (44.6)	91.3%	98.3%	32.5%	85.8%
12 mm	54.4 (57.9)	**98.4% (Pass)***	99.8%	44.6% (Fail)	**95.6% (Pass)***

Figure 3: 3D Marching Cubes Surface Rendering of Thoracic Structures and Target Excursion

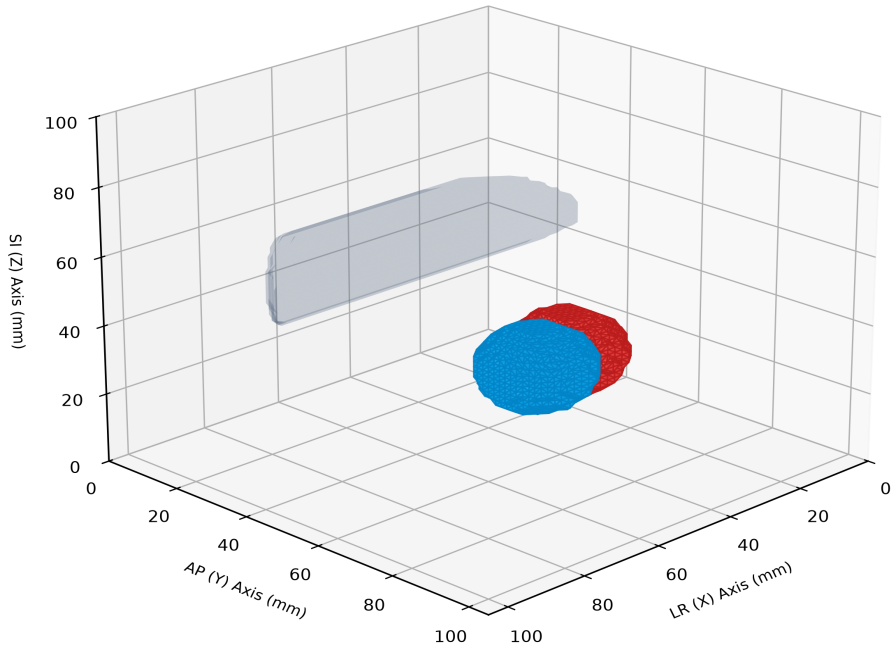


Figure 3. 3D Marching Cubes surface extraction of thoracic anatomy and target excursion.

4. Discussion

4.1. Kinematic Agreement with Clinical Motion Cohorts:

Our computational results confirmed substantial directional asymmetry, with longitudinal (SI) translation vastly exceeding lateral (LR) and anteroposterior (AP) motion. The simulated SI amplitudes (12.0 mm to 22.0 mm) agree closely with physical patient measurements reported in the radiotherapy literature. Specifically, clinical fluoroscopic evaluations of lower-lobe lung lesions documented by Barnes et al. [18] identified a mean craniocaudal excursion of 18.5 mm (observed cohort range: 9 to 32 mm). In addition, clinical studies by Erridge et al. [19] and Hanley et al. [20] recorded mean SI displacements of 12.5 mm and 12.0 mm, respectively. The modest transverse and sagittal excursions obtained in our pipeline (1.18 to 6.07 mm) also reflect the restricted mechanical mobility observed clinically by Seppenwoolde et al. (mean LR = 1.5 mm, AP = 2.5 mm) [21].

4.2. Geometric Limitations of Symmetric Buffers:

In current clinical protocols, uniform 5 mm margins are often prescribed based on static setup tolerance. However, our spatiotemporal analysis reveals that a 5 mm margin provides only 59.3% target enclosure during steady respiration and falls to 9.5% during deep breathing, indicating that over 90% of the lesion volume escapes the target envelope during peak inhalation. While an expanded 12 mm buffer satisfies the 95% threshold for steady cycles (CR = 98.4%), it increases the irradiated volume by +444% (from 10.0 cc to 54.4 cc). In thoracic treatment plans, such volumetric increases significantly raise the mean lung dose and V20Gy parameter, directly correlating with clinical incidence of radiation pneumonitis [22, 23].

4.3. Recommendations for Clinical Motion Management:

Because lung tumor excursion is fundamentally directional, uniform margin expansion unnecessarily exposes healthy tissues along the lateral and sagittal planes while failing to secure target coverage along the craniocaudal trajectory. Clinical planners should therefore prioritize: (1) Directional anisotropic margins tailored to patient-specific trajectory

amplitudes; or (2) Active motion management protocols (such as Deep Inspiration Breath Hold - DIBH or phase-based respiratory gating) as mandated by AAPM TG-76 for any patient exhibiting SI excursions greater than 5 mm [1, 24, 25].

4.4. Study Limitations

Several methodological limitations must be considered when interpreting these findings:

1. **Geometric Enclosure versus Dosimetric Accumulation:** Target coverage in this study was evaluated based on binary volumetric intersection (CR%). In clinical delivery, radiation dose distributions in heterogeneous lung parenchyma are governed by lateral electronic disequilibrium. Full evaluation requires Monte Carlo or Collapsed Cone Convolution 4D dose accumulation to determine the biological Effective Uniform Dose (EUD) [26].
2. **Rigid Target Assumption:** The tumor was modeled as a rigid spherical volume. Clinical thoracic lesions often undergo non-rigid deformation, shearing, and volumetric expansion/contraction during breathing, which necessitates deformable image registration (DIR) algorithms for precise mapping [27].
3. **Interfractional Motion Variability:** Our model evaluates intrafractional motion within a single 4D-CT cycle. In multi-week fractionated radiotherapy, day-to-day baseline shifts, patient posture changes, and radiation-induced tumor shrinkage introduce interfractional uncertainties that require adaptive replanning [28].
4. **Surrogate Hysteresis:** Real-world external respiratory surrogates (such as RPM chest blocks or spirometry) often exhibit time lags and non-linear phase shifts relative to internal tumor motion (hysteresis loops), which introduce residual motion uncertainties during prospective gating [1, 21].

5. Conclusion

This study presented an open-source Python computational framework for spatiotemporal 4D-CT tumor tracking and margin assessment. Quantitative results demonstrate that uniform 5 mm margins fail to maintain adequate target coverage under dynamic respiration. Directional margin design and active respiratory management are essential to ensure target coverage while sparing normal pulmonary tissue.

Data and Code Availability Statement

The complete Python simulation pipeline, synthetic 4D-CT digital phantom arrays, and margin analysis scripts developed in this study are publicly available on GitHub at https://github.com/anhvu7123/4d-ct-lung-motion under the open-source MIT License, and permanently archived on Zenodo at https://doi.org/10.5281/zenodo.10892026. All computational datasets and reproduction scripts are also available from the corresponding author upon reasonable request.

Declaration of Generative AI and AI-assisted Technologies

During the preparation of this work, the author used generative AI tools to assist with English language polishing and code debugging. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT Authorship Contribution Statement

Vu Minh Anh: Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interests

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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